VERSION
14.0

AMS GRA ARCHITECTURE VOLUMES

Open Mission Systems (OMS) Service MPU

AFLCMC/WA

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1.0 Overview

An Open Mission Systems (OMS) Service is a Level 1 Minimum Procurable Unit (MPU) in the Agile Mission Suite (AMS) Government Reference Architecture (GRA). An OMS Service is an OMS Architecture Element that implements a set of behaviors and functions accessed In Accordance With (IAW) a defined open interface using OMS Messages by utilizing the Abstract Service Bus (ASB). A non-proprietary service contract describes the OMS Service. OMS Services are deployed on Digital Payload Processors (DPPs) in the Compute Cluster Farm (CCF).

1.1 Purpose

The OMS Service MPU Volume introduces the OMS Service MPU and describes its key features, logical context, interfaces, and compliance requirements. The key features (**Table 2.1.1**) provide design guidance on what is typically expected for the OMS Service MPU. The interface diagram (**Figure 2.2.1**) contains a graphical representation of the types of interfaces that can exist, from the perspective of the OMS Service MPU. The interface table (**Table 2.3.1**) contains a textual representation of the interfaces and their descriptions. Compliance requirements (**Table 3.0.1**) describe what is required for the OMS Service MPU to achieve an AMS GRA compliance tier.

1.2 Scope

This OMS Service MPU Volume supports compliance assessment for AMS GRA adopters.

1.3 Authoritative References

Authoritative references are key documents that a procuring authority, Mission System Integrator (MSI), or MPU provider refers to when determining whether a given AMS GRA entity is compliant with AMS GRA business and architecture requirements. An AMS GRA entity is compliant when it provides the required prescriptive constructs (e.g. interfaces for an MPU). See **Table 1.3.1** for a list of authoritative references for the OMS Service MPU Volume.

Standard	Title	Version/Date
AMS GRA Volumes	Compliance Assessment Approach Volume	14.0
AMS GRA Volumes	Mission System Volume	14.0
OMS Standard	OMS Standard	Version identified by the MSI
AMS GRA Volumes	OMS Service Contract Addendum Template	14.0

AMS GRA Volumes	CDS IDD	14.0

Table 1.3.1: Authoritative References Identify the Key Documents Necessary for Performing a Compliance Assessment of OMS Service MPU.

1.4 Informative References

Informative references are supplemental documents that a procuring authority, MSI, or MPU provider refers to for more information about an AMS GRA entity or related entities. See **Table 1.4.1** for a list of informative references for the OMS Service MPU Volume.

Standard	Title	Version/Date
AMS GRA Volumes	Software Architecture Volume	14.0

Table 1.4.1: Informative References Identify Documents a Procuring Authority, MSI, or MPU Provider Can Use for More Information About the OMS Service MPU and Related Entities.

2.0 Open Mission Systems (OMS) Service

OMS Services provide a standardized approach for software deployment. These services are a core element of OMS, the United States Air Force (USAF) Mission System Software standard. OMS Services are intended for software deployed on a wide variety of platforms. OMS Services can be mission-agnostic or mission specific. The Mission Agnostic Service Infrastructure (MASI) includes OMS Services.

Leveraging OMS for its software Services enables future platforms to take advantage of existing and matured OMS software Services. OMS Services are built for a DPP Compute Environment. These Services use the DPP Application Programming Interface (API) to access various facilities provided by the DPP hardware and its operating system/environment. As an example, an OMS Service may access the DPP system time through the DPP API. OMS Services are typically deployed on General Purpose Processor (GPP) components within a DPP.

OMS implements common message interfaces. These interfaces enable an OMS Service to be replaceable and competitive, while reducing the integration cost between the subsystem and Service levels. OMS Services must communicate by utilizing standard messaging on the ASB using the OMS Critical Abstraction Layer. OMS Critical Abstraction Layer implementations tend to be complex, with non-trivial computing resource requirements and message latencies. The OMS Critical Abstraction Layer message evolution is constrained by the governance of a common message schema.

2.1 Key Features

Table 2.1.1 identifies the key features of OMS Services and the benefit of these features. This table is not prescriptive. However, the table identifies a number of the features and attributes that AMS GRA interfaces are designed to enable. This table is also included in the OMS Service documentation template so that OMS Service developers can summarize features supported by specific OMS Service instantiations.

Key Feature	Benefit
Deployed on standardized GPP components in DPP MPUs	Standardized approach for general-purpose software deployment
Interfaces to a Service are Common	Replaceability, competition, reduced integration cost
Service behaviors and use of interfaces are defined in a non-proprietary Service contract	Replaceability, competition, reduced integration cost

Key Feature	Benefit
Services access most interfaces through OCE-provided APIs	Increased portability of software across heterogeneous CCF hardware
Services are built using tools provided by each DPP DK	Increased portability of software across heterogeneous CCF hardware
Status Reporting	OMS Services report Service Status IAW the OMS standard. This standard Service Status enables mission system management software to monitor Services in a common way
Standardized Marketplace	Provides standardized application store to pull Services per adopter and platform needs

Table 2.1.1: The OMS Service Key Features Table Describes the Specific Features Supported by an OMS Service Instantiation.

2.2 Interface Diagram

Figure 2.2.1 depicts the OMS Service and the corresponding execution environment provided by a DPP MPU. The interface diagram only contains logical entities. Interface connections denoted as pass-through only represent situations where the sending MPU is not the original source and does not use or modify what is provided to the receiving MPU.

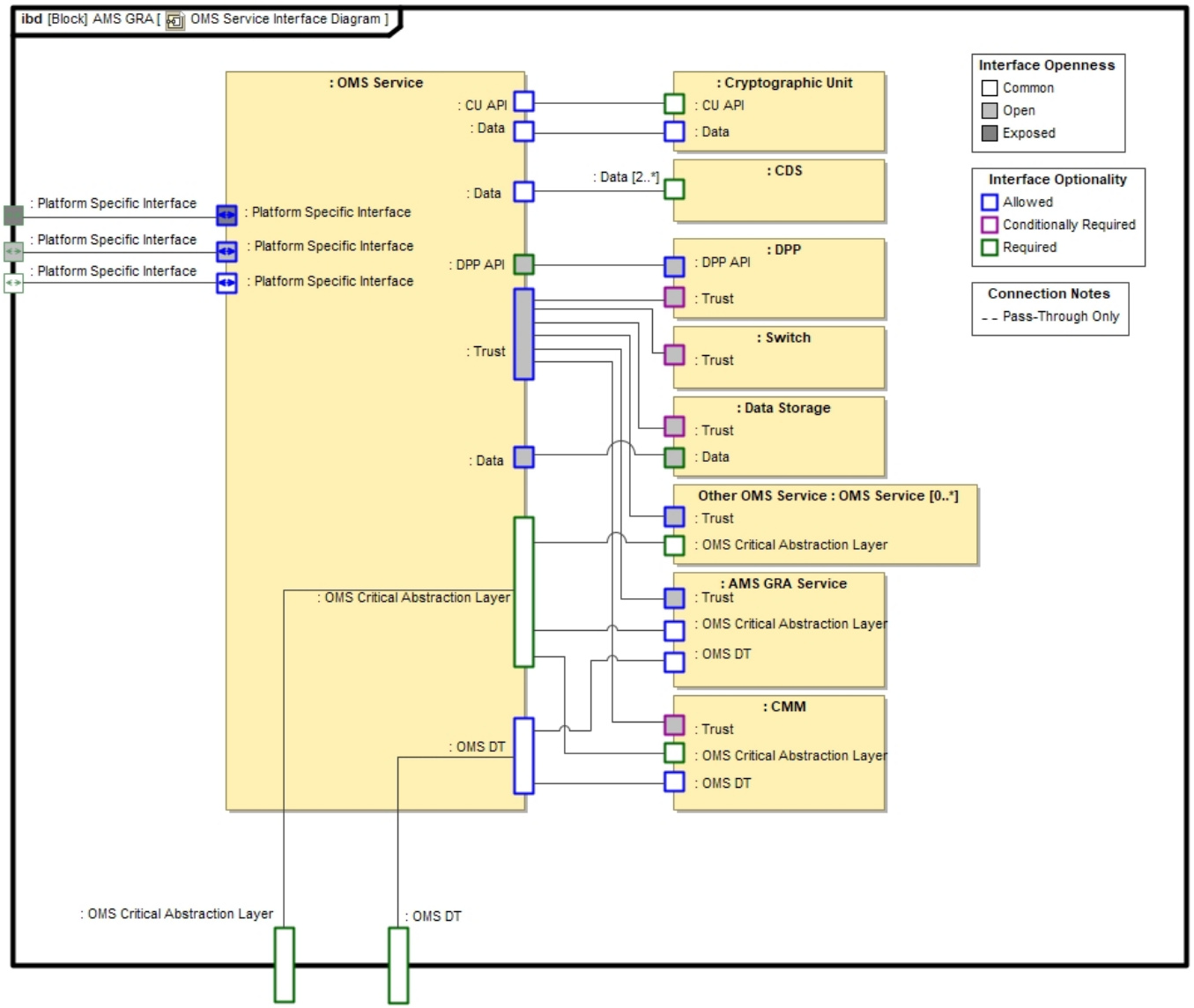


Figure 2.2.1: This OMS Service MPU Interface Diagram Shows Logical Interfaces of the OMS Service MPU and Their Connections with Other AMS GRA MPUs.

2.3 Interfaces

Table 2.3.1 identifies the OMS Service interfaces depicted in **Figure 2.2.1**. When determining whether a given MPU or infrastructure is AMS GRA compliant, Interface Optionality is defined as follows:

- Required interfaces must be implemented.
- Conditionally Required interfaces must be implemented when specific conditions are satisfied.
- Allowed interfaces are optional to implement.

For interfaces listed multiple times, each instance represents an openness category that an interface could belong to. The Platform Specific Interface can fall into any of the three openness categories because some MPUs may instantiate an allowable interface at one or more openness levels. Those interfaces will be represented on the corresponding diagram and table for each allowable openness level.

Interface	Documentation	Interface Optionality
Platform Specific Interface	The Common Platform Specific Interface provides an information flow between OMS Service and non-proprietary hardware and software that follow AMS GRA standards.	Allowed
Platform Specific Interface	The contractually agreed upon Exposed Platform Specific Interface provides an information flow between OMS Services and proprietary hardware and software.	Allowed
Platform Specific Interface	The Open Platform Specific Interface provides an information flow between OMS Service and non-proprietary hardware and software not required to follow AMS GRA standards.	Allowed
Trust	The Trust interface enables secure connectivity between endpoints using cryptographic means to establish a secure channel. This channel may be used to verify integrity, authenticate endpoints, or to transfer data requiring higher levels of assurance than normal data transfers, such as Critical Program Information (CPI), or Software. The implementation of this interface may vary depending on assurance level, exportability, and the various mission specific requirements of the adopting program.	Allowed
Data	The Data interface specifies the data interfaces between the OMS Service and other MPUs.	Allowed
DPP API	The DPP API provides the Service Level 1 MPUs with access to various facilities provided by the DPP Compute Environment. Note that the GPP variant of the DPP covers function calls for services to be able to query for Time as distributed by the Mission System.	Required

Interface	Documentation	Interface Optionality
OMS Critical Abstraction Layer	The Critical Abstraction Layer API provides access to the ASB. The ASB is a logical construct that includes OMS Critical Abstraction Layer messaging, data transfer, and security messages. See the OMS standard for additional information. The OMS Service is required to use an implementation of the Critical Abstraction Layer API per OMS Definition and Documentation.	Required
OMS DT	The OMS Data Transfer interface provides access to the OMS Internet Protocol-based messages when appropriate.	Allowed
CU API	The Cryptographic Unit (CU) API enables AMS GRA Service to access the CU's cryptographic functionalities.	Allowed

Table 2.3.1: The Procuring Authority, MPU Provider, and MSI Use this Table to Identify the Interfaces to Implement for a Compliant MPU or Infrastructure.

3.0 Compliance Requirements

Table 3.0.1 gives the applicable MPU compliance requirements for the OMS Service MPU.

See the Compliance Assessment Approach Volume for more information on the tiered requirement strategy as well as the definitions of the interface characteristics.

See the Mission System Volume for additional details about MPU Fault Reporting.

MPU developers need to coordinate with the procuring authority and applicable MSI to receive additional guidance beyond basic compliance. These compliance requirements include architectural guidance to ensure the MPU integrates within the overall Mission System architecture as well as guidance to align with Commercial-Off-The-Shelf (COTS) standards (e.g., physical form factor, cooling strategy, environmental operating conditions, etc.).

For OMS Service MPUs that provide OMS adapter functionality, compliance is defined in the AMS GRA Compliance Assessment Approach, Section 2.1 Basic Rules of Compliance.

For OMS Service MPUs that provide OMS isolator functionality, compliance is defined by adherence to GRA-MPU-002, GRA-MPU-004, GRA-MPU-026, and GRA-MPU-027.

Note: The verification methods listed in the requirements table are AMS GRA recommendations. An adopting program may alter the recommended verification methods IAW the needs of the program. Refer to the Compliance Assessment Approach Volume for more information.

Id	Text	Compli-ance Tier	Recommended Verification Method	Notes
GRA-MPU-002	An MPU shall be documented IAW the OMS Service Contract Template and OMS Service Contract Addendum.	Tier 1	Inspection	
GRA-MPU-003	An MPU targeting deployment on a GPP DPP shall be restricted to using the POSIX API for OS system calls.	Tier 1	Inspection	
GRA-MPU-004	An MPU shall meet all of the OMS Tier 1 requirements for an OMS Service.	Tier 1	Inspection	
GRA-MPU-005	An MPU with CPI on a Trusted Ops platform shall cryptographically protect CPI using an open implementation.	Tier 3	Analysis	
GRA-MPU-006	An MPU that processes or decrypts CPI on a built-for-export platform shall implement the OMS Standard.	Tier 2	Analysis	
GRA-MPU-024	An MPU that interacts with the CDS or Cryptographic Unit shall use the CDS IDD or CU IDD respectively interface for all exchanges with the Cryptographic Unit or CDS.	Tier 1	Inspection	
GRA-MPU-025	An MPU shall report faults with the associated fault status.	Tier 1	Inspection	Off-nominal behavior detailed in compliance documentation.
GRA-MPU-026	An MPU shall implement Required interfaces specified IAW the AMS GRA Model.	Tier 1	Inspection	
GRA-MPU-027	An MPU shall implement all interfaces with the level of Openness specified IAW the AMS GRA Model.	Tier 1	Analysis	
GRA-MPU-030	An MPU shall document interface openness IAW the Compliance Assessment Approach.	Tier 1	Inspection	
GRA-MPU-031	An MPU shall include a model compatible with MBSE content contained in the AMS GRA SK release with which the MPU is compliant.	Tier 2	Inspection	
GRA-MPU-032	An MPU shall include a model that identifies all external MPU interfaces.	Tier 2	Inspection	
GRA-MPU-039	An MPU shall provide a CLASSIFICATION.md file defining security classification of included file contents.	Tier 2	Inspection	
GRA-MPU-040	An MPU shall provide a DATARIGHTS.md file defining data rights of included file contents.	Tier 2	Inspection	
GRA-MPU-041	An MPU shall document SAST results for all microprocessor software source code in the MPU, IAW the MPU documentation template.	Tier 1	Inspection	

Id	Text	Compli-ance Tier	Recommended Verification Method	Notes
GRA-MPU-042	An MPU shall document DAST results for all microprocessor software in the MPU, IAW the MPU documentation template.	Tier 1	Inspection	
GRA-MPU-046	An MPU shall provide automated tests of non-COTS microprocessor software with at least 50% source code structural coverage (excluding third-party COTS or open source libraries).	Tier 1	Demonstration	Only applicable to MPUs that include non-COTS microprocessor code.
GRA-MPU-047	An MPU with microprocessor code (excluding third-party COTS or open source libraries) shall only use relative file paths in the execution environment.	Tier 2	Inspection	
GRA-MPU-048	An MPU with microprocessor code (excluding third-party COTS or open source libraries) shall only use relative file paths in the build environment.	Tier 2	Inspection	
GRA-MPU-051	An MPU shall include a software bill of materials in an open and non-proprietary format.	Tier 1	Inspection	
GRA-MPU-053	An MPU's DPs shall each contain content only from that MPU.	Tier 3	Inspection	
GRA-MPU-057	An MPU's DP that targets a GPP Processing Component shall consist of a single OCI-compliant container.	Tier 1	Demonstration	Note: An MPU that is delivered in a container is allowed to be removed from the container for execution. Services may also be delivered in a non-containerized format in addition to delivery in a containerized format.
GRA-MPU-058	An MPU shall encrypt classified DaR using NSA-approved algorithms.	Tier 1	Analysis	
GRA-MPU-059	An MPU that decrypts and/or encrypts classified data and/or CPI shall only store classified plain text keys in volatile memory.	Tier 1	Inspection	
GRA-MPU-066	An MPU shall support the use of a Mission System-provided RoT.	Tier 1	Demonstration	
GRA-MPU-071	An MPU shall be built, packaged, and delivered to persistent storage via automated pipeline.	Tier 1	Inspection	
GRA-MPU-074	An MPU that processes or decrypts CPI on a built-for-export platform shall fully implement OMS Standard.	Tier 3	Analysis	

Id	Text	Compli-ance Tier	Recommended Verification Method	Notes
GRA-MPU-077	An MPU shall provide automated tests of non-COTS microprocessor software with at least 80% source code structural coverage (excluding third-party COTS or open source libraries).	Tier 3	Demonstration	
GRA-MPU-078	An MPU model shall be capable of automated verification with AMS GRA models.	Tier 3	Inspection	
GRA-MPU-079	An MPU shall include a model that provides definition of all external MPU interfaces.	Tier 3	Inspection	
GRA-MPU-082	An MPU shall pass 70% or more of applicable jobs in a AMS GRA compliant pipeline.	Tier 2	Inspection	
GRA-MPU-083	An MPU shall pass 100% of applicable jobs in a AMS GRA compliant pipeline.	Tier 3	Inspection	
GRA-MPU-085	An MPU shall report faults with the associated fault status using the applicable interface. (See notes).	Tier 2	Inspection	Cryptographic Unit MPUs and CDS MPUs report faults as defined in the CDS and CU IDD. Local Function MPUs report faults as defined in Local Function DK. MFA MPUs report faults as defined in the applicable MEL IDD. AMS GRA Service, CMM, DPP, Data Storage, Switch, OMS Service, and PCaB MPUs report faults using a software interface in UCI format.
GRA-MPU-086	An MPU shall use the CMM IDD for all information exchanges between the MPU and the CMM.	Tier 1	Inspection	
GRA-MPU-090	An MPU shall provide a configuration file named service.yml with contents in accordance with the Service MPU Configuration Template YAML file.	Tier 1	Inspection	

Table 3.0.1: An OMS Service's Adherence to the Compliance Requirements Determines Its Compliance Score. See the Compliance Assessment Approach Volume for More Details.

A.0 Acronyms

AMS	Agile Mission Suite
API	Application Programming Interface
ASB	Abstract Service Bus
CCF	Compute Cluster Farm
CDS	Cross Domain Solution
CMM	Countermeasures Manager
COTS	Commercial-Off-The-Shelf
CPI	Critical Program Information
CU	Cryptographic Unit
DaR	Data at Rest
DAST	Dynamic Application Security Testing
DK	Development Kit
DP	Digital Payload
DPP	Digital Payload Processor
GPP	General Purpose Processor
GRA	Government Reference Architecture
IAW	In Accordance With
IDD	Interface Description Document
MASI	Mission Agnostic Service Infrastructure
MBSE	Model-Based Systems Engineering
MEL	Multi-Function Aperture (MFA) Encapsulation Layer
MFA	Multi-Function Aperture
MPU	Minimum Procurable Unit
MSI	Mission System Integrator
NSA	National Security Agency
OCE	Open Compute Environment
OCI	Open Container Initiative

OMS	Open Mission Systems
PCaB	Processor Chassis and Backplane
RoT	Root of Trust
SAST	Static Application Security Testing
SK	Starter Kit
UCI	Universal Command and Control Interface
USAF	United States Air Force
YAML	YAML Ain't Markup Language

B.0 Revision History

The Revision History table, **Table B.0.1**, captures the major changes made for a given AMS GRA volume as the Starter Kit (SK) continues to mature.

Revision	Description	Date	Notes
-	This revision adds a revision table to the volume.	May 22, 2024	SK11.0
A	This revision did not introduce any significant changes.	July 24, 2025	SK13.0
B	This revision removes references to the security module. Updates GRA-MPU-003, GRA-MPU-004, GRA-MPU-051, and GRA-MPU-071 for clarity. Adds new requirement GRA-MPU-090.	April 24, 2026	AMS GRA 14.0

Table B.0.1: Revision History Captures the Major Changes Made to the OMS Service MPU Volume Between Documentation Releases.